WORKSHEET 0.4

Your Graphing Calculator: The Four Exam Skills

Unit 0: Readiness

Key ideas

- The exam expects four calculator skills: **graph** in a window, **solve** an equation, find a **derivative at a point**, and evaluate a **definite integral**.
- Work in **radian mode**. Give final answers correct to **three decimal places**.
- **Store** intermediate values at full precision; round once, at the end.
- Write the set-up in standard notation (an equation, $f'(a)$, or an integral with limits) *before* the decimal. Calculator syntax earns nothing.

Worked examples

Example 1 · Solving an equation

Find the positive solution of $x^4 - x - 3 = 0$.

Solution. Graph $y = x^4 - x - 3$; it crosses the positive x-axis once, between 1 and 2. Use the zero command.

$x^4 - x - 3 = 0$ **when** $x \approx 1.453$**.**

Example 2 · A derivative at a point

Let $f(x) = x^2 \ln x$. Find $f'(1.7)$.

Solution. Use the numerical derivative command at $x = 1.7$.

$f'(1.7) \approx 3.504$**.**

Example 3 · A definite integral

Evaluate $\displaystyle\int_1^2 \sqrt{x^3 + 1}\, dx$.

Solution. Use the numerical integral command with limits 1 and 2.

$\displaystyle\int_1^2 \sqrt{x^3 + 1}\, dx \approx 2.130$**.**

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Practice

A graphing calculator is required for every question on this page. Work in radian mode and give decimal answers to three decimal places.

CORE

1. Find the smaller of the two solutions of $e^x = 4x$.

2. Find the value of x at which the graphs of $y = x^3$ and $y = \cos x$ intersect.

3. Let $f(x) = \arctan(x^2)$. Find $f'(1.3)$.

4. Evaluate $\displaystyle\int_0^1 e^{x^2}\,dx$.

EXAM LEVEL

5. Find the larger of the two solutions of $e^x = 4x$.

6. The graphs of $y = 4 - x^2$ and $y = e^x$ intersect at $x = a$ and $x = b$, with $a < b$. Find a and b, and find $\displaystyle\int_a^b \left(4 - x^2 - e^x\right) dx$.

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7. Let $f(x) = \sin(e^x)$. Find $f'(0.5)$.

8. A student writes "fnInt(X sin(X), X, 0, π) = 3.14". Rewrite this line so that it would earn full credit.

Question 9 · Free-response practice

Let $f(x) = 3\ln(x+1)$ and $g(x) = x^2$. The graphs of f and g intersect at $x = 0$ and at $x = c$, where $c > 0$.

(a) Find c. Write the equation you solved.

(b) Find the area of the region enclosed by the two graphs. Write the integral before the decimal.

(c) Find $f'(1)$ and $g'(1)$. Which graph is steeper at $x = 1$?

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Worked solutions

1. $e^x - 4x = 0$ when $x \approx 0.357$.
2. $x^3 = \cos x$ when $x \approx 0.865$.
3. $f'(1.3) \approx 0.674$.
4. $\displaystyle\int_0^1 e^{x^2}\,dx \approx 1.463$.
5. $x \approx 2.153$.
6. $a \approx -1.965$ and $b \approx 1.058$ (stored at full precision). $\displaystyle\int_a^b \left(4 - x^2 - e^x\right) dx \approx 6.428$.
7. $f'(0.5) \approx -0.128$.
8. $\displaystyle\int_0^\pi x \sin x\,dx \approx 3.142$. Standard notation, and three decimal places.
9. (a) $3\ln(c+1) = c^2$ gives $c \approx 1.738$.
   (b) $\displaystyle\int_0^c \left[3\ln(x+1) - x^2\right] dx \approx 1.309$.
   (c) $f'(1) = 1.5$ and $g'(1) = 2$, so the graph of g is steeper at $x = 1$.

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